

Billiard

Tags:

Time Limit: 1 s Memory Limit: 128 MB
 Submission: 39 AC: 22 Score: 99.81

Submit

Codes

AI Code Tips

Description

Problem A - Billiard

In a billiard table with horizontal side **a** inches and vertical side **b** inches, a ball is launched from the middle of the table. After **s** > 0 seconds the ball returns to the point from which it was launched, after having made **m** bounces off the vertical sides and **n** bounces off the horizontal sides of the table. Find the launching angle **A** (measured from the horizontal), which will be between 0 and 90 degrees inclusive, and the initial velocity of the ball.

Assume that the collisions with a side are elastic (no energy loss), and thus the velocity component of the ball parallel to each side remains unchanged. Also, assume the ball has a radius of zero. Remember that, unlike pool tables, billiard tables have no pockets.

Input

Input consists of a sequence of lines, each containing five nonnegative integers separated by whitespace. The five numbers are: a, b, s, m, and n, respectively. All numbers are positive integers not greater than 10000. Input is terminated by a line containing five zeroes.

Output

For each input line except the last, output a line containing two real numbers (accurate to two decimal places) separated by a single space. The first number is the measure of the angle A in degrees and the second is the velocity of the ball measured in inches per second, according to the description above.

Samples

input	Copy
100 100 1 1 1 200 100 5 3 4 201 132 48 1900 156 0 0 0 0 0	
output	Copy
45.00 141.42 33.69 144.22	

3.09 7967.81

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 15:10:41